Overview of Ecosystem Service: Coastal Carbon Storage

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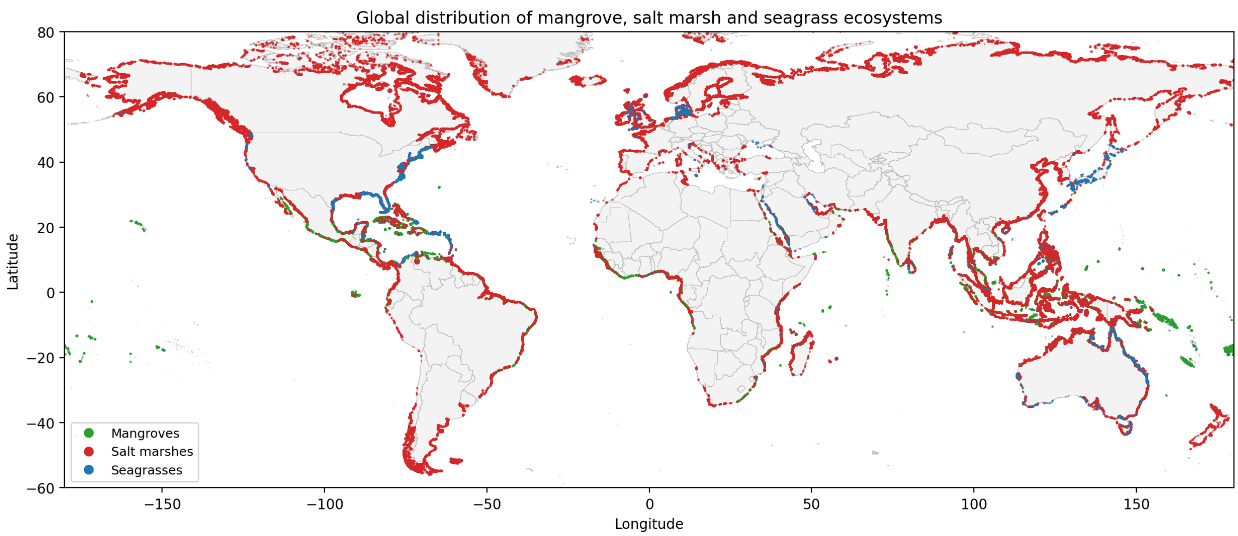
1 Introduction

Mangroves, salt marshes and seagrass meadows store large amounts of organic carbon and provide an important blue carbon ecosystem (BCE) service for climate change mitigation (McLeod et al. 2011; Howard et al. 2014; IPCC 2014; Macreadie et al. 2021) (Figure 1). This report estimates the country-level economic value of the carbon stock held in these three BCE types within the Gross Ecosystem Product (GEP) framework. We assign stocks to countries using each coastal country's marine Exclusive Economic Zone (EEZ), following the EEZ-based attribution used by Bertram et al. (Bertram et al. 2021), and value the standing stock with the rental social cost of carbon.

Figure 1: Global distribution of mangrove, salt marsh and seagrass ecosystems



Notes: This figure uses Global Mangrove Watch v3 (Bunting et al. 2022), the GLC_FCS30D salt marsh layer (Zhang et al. 2024) and the UNEP-WCMC Global Distribution of Seagrasses v7.1 (UNEP-WCMC and Short 2021). The three-ecosystem framing follows Macreadie et al. (Macreadie et al. 2021).

2 Data and Methods

The spatial domain is each coastal country's marine Exclusive Economic Zone (EEZ). We use the EEZ as the country-attribution unit for coastal carbon, following Bertram et al. (Bertram et al. 2021). This domain captures the coastal-water envelope where mangroves, salt marshes and seagrasses occur. It also keeps the coastal-carbon account conceptually separate from the terrestrial carbon service reported in another section of the Global GEP account.

2.1 Data

Mangrove extent comes from the 2019 Global Mangrove Watch v3 vector product (Bunting et al. 2022), which maps mangrove forests globally at 30 m using ALOS PALSAR and Landsat observations. Salt marsh extent comes from a pre-processed vector layer extracted from the 2019 Zhang et al. GLC_FCS30D global land-cover product (Zhang et al. 2024). Seagrass extent comes from the UNEP-WCMC Global Distribution of Seagrasses v7.1 polygon product (UNEP-WCMC and Short 2021), a 2021 compilation of 293,147 polygons based on field observations from the late nineteenth century to 2018. No wall-to-wall global seagrass map exists for 2019, so we use WCMC v7.1 as the best available spatial baseline and treat the temporal mismatch as a major uncertainty. Its area totals fall within the 160,000 to 267,000 km² global extent range synthesized by McKenzie et al. (McKenzie et al. 2020), which is also the extent basis used by Gomis et al. (Gomis et al. 2025).

Mangrove soil organic carbon comes from the Sanderman et al. 30 m random-forest map for the top 20 cm of soil (Sanderman et al. 2018), scaled from the original 1 m product assuming a linear depth profile. We use the median-typology product for 2019 to 2020, reproject it once to the 10 arc-second grid using average resampling, and cache the result for later runs. Salt marsh soil carbon uses a latitude step function based on Maxwell et al. (Maxwell et al. 2024), scaled to 20 cm depth, with values rising from ~36 ton C/ha in the tropics to ~70 ton C/ha at high latitudes. Seagrass soil carbon uses the Fourqurean et al. (Fourqurean et al. 2012) global mean of 22 ton C/ha for the top 20 cm.

2.2 Quantifying the blue carbon stock

The country-level one-year storage value for 2019 has two steps. First, we estimate the standing organic carbon stock in each coastal country's EEZ. Second, we convert that stock into an annual monetary value for one year in 2019. For each ecosystem, carbon stock is the sum of above-ground biomass carbon, below-ground biomass carbon and soil organic carbon to 20 cm depth.

2.2.1 Mangroves

Above-ground biomass density uses the Hamilton and Friess (Hamilton and Friess 2018) combined latitude regression, $AGB = \max(0, -6.4305\,|\mathrm{lat}| + 271.747)$ in tons per hectare. This regression combines the earlier Twilley et al. (Twilley, Chen, and Hargis 1992) and Saenger and Snedaker (Saenger and Snedaker 1993) equations. We convert AGB to carbon using a factor of 0.48 (Hamilton and Friess 2018).

Below-ground biomass is allocated from AGB using the IPCC Wetlands Supplement Table 4.6 climate-zone ratios (IPCC 2014). Tropical wet pixels (`|lat| < 23.5°` and mean annual precipitation greater than 2,000 mm) use a BGB:AGB ratio of 0.49. Tropical dry pixels use 0.29, and subtropical pixels (`|lat|   23.5°`) use 0.96. If no precipitation raster is configured, all tropical pixels default to the wet-tropical ratio. This is reasonable for most global mangroves, but it can overestimate BGB along dry-tropical coasts such as Arabia and northern Australia. We convert BGB to carbon using a factor of 0.39 (Howard et al. 2014).

Soil organic carbon is read directly from the Sanderman map at the pixel scale (scaled to 20 cm). If the Sanderman raster is not available locally, the pipeline uses a fallback latitude step function that decreases from 80 ton C/ha at the equator to 50 ton C/ha by 25° latitude. For example, a tropical wet equatorial pixel with Sanderman SOC = 80 ton C/ha (scaled from 1 m to 20 cm) has about 130 + 52 + 80   262 ton C/ha total ecosystem carbon to 20 cm depth.

2.2.2 Salt marshes

Salt marsh above-ground biomass is set to a global median of 5 t/ha (Howard et al. 2014). For tropical marshes (`|lat| < 25°`), we increase this value by 20 percent because warm, frequently flooded marshes tend to be more productive. We estimate below-ground biomass as 2.5 times above-ground biomass, which falls within the published 2:1 to 5:1 range. We convert biomass to carbon using factors of 0.45 for AGB and 0.41 for BGB (Howard et al. 2014).

Salt marsh soil organic carbon uses a four-bin latitude step function based on Maxwell et al. (Maxwell et al. 2024), scaled to 20 cm: 36 ton C/ha in the tropics, 44 ton C/ha in the warm temperate zone, 50 ton C/ha in the cool temperate zone, and 70 ton C/ha in peat-rich boreal and arctic marshes. A native raster from that source is not yet wired into the pipeline, so this step function captures the broad latitude pattern but not the large within-zone variability in the source dataset.A representative salt marsh pixel at 30°N with median productivity and 44 ton C/ha soil carbon yields about 2.25 + 5.13 + 44 51 ton C/ha total to 20 cm depth. Most salt marsh carbon is therefore stored in soil.

2.2.3 Seagrasses

Seagrass biomass carbon densities are assigned by polygon using genus-specific means from Gomis, Strydom, Foster et al. (Gomis et al. 2025). Total biomass carbon ranges from 9.78 ton C/ha for *Posidonia* to 0.19 ton C/ha for *Halophila*. Other persistent genera (*Phyllospadix*, *Enhalus*, *Cymodocea*, *Thalassia*, *Amphibolis*, *Thalassodendron*) use ~3.70 ton C/ha; opportunistic genera (*Zostera*, *Syringodium*, *Halodule*) use ~0.94 ton C/ha; and *Ruppia* uses 0.33 ton C/ha. Polygons without genus identification, about 78 percent of the WCMC dataset, receive the Gomis global mean of 1.55 ton C/ha. We split total biomass into 30 percent above-ground and 70 percent below-ground, reflecting the dominance of roots and rhizomes in seagrass biomass (Howard et al. 2014; Fourqurean et al. 2012). Freshwater plant genera (*Trapa*, *Myriophyllum*, *Valisneria*, *Najas*, *Heteranthera*, *Stuckenia*, *Utricularia*) are set to zero before aggregation.

Seagrass soil organic carbon is set to 22 ton C/ha for the top 20 cm, the direct Fourqurean et al. (Fourqurean et al. 2012) global mean. This value is applied uniformly to all polygons, so the current model does not capture spatial heterogeneity in seagrass soil carbon. Empirically, seagrass soil carbon varies substantially by genus and environmental setting: *Posidonia* meadows in Mediterranean anoxic basins can exceed 100 ton C/ha, while *Halophila* meadows in clear, high-energy settings may contain only 2.5 to 7.5 ton C/ha. The global mean of 22 ton C/ha is a reasonable central estimate but masks this variation. A conservative low-carbon scenario (11 ton C/ha) would reflect slower-growing, shorter-lived genera and shallow, well-oxygenated seagrass meadows; a high-carbon scenario (60 ton C/ha) would reflect peat-rich, long-lived meadows in low-energy lagoons and estuaries. We quantify the sensitivity of global value to this uncertainty in the Results section.

2.3 Pricing the carbon stock with the rental SCC

We value the standing coastal carbon stock with the rental social cost of carbon (SCC) (Parisa et al. 2022; Groom and Venmans 2023; Venmans, Rickels, and Groom 2025). The rental SCC measures the avoided climate damage from keeping one unit of carbon out of the atmosphere for one additional year. Within the GEP framework, we estimate the annual rental value of standing-stock storage as

$$\text{rental SCC}(t) = P_C(t) - P_C(t+1)\, e^{-r} = P_C(t) - P_C(t)\, e^{g}\, e^{-r} = P_C(t)\,(1 - e^{-x}),$$

where $P_C(t)$ is the social cost of carbon in year t, $P_C(t+1)$ is the social cost of carbon in year $t+1$, r is the discount rate, g is the SCC growth rate, and $x = r - g$ is the net carbon discount rate. The rental SCC is the annual benefit of storing one metric ton of CO for one year. The country-level value reported here is the one-year 2019 storage value:

$$GEP_{\text{coastal carbon}} = C_{\text{total}} \times \text{rental SCC} \times 1.$$

The base year is 2019, so the one-year rental rate requires $P_C(2019)$ and $P_C(2020)$. The Greenhouse gas Impact Value Estimator (GIVE) model begins in 2020, so we approximate $P_C(2019)$ using $P_C(2020)$. GIVE is used because it underpins the updated U.S. EPA social cost of carbon and combines four damage channels: mortality, agriculture, energy and sea level rise (Rennert et al. 2022). For the baseline rental calculation, we assume a constant real SCC over the one-year interval and set $g = 0$. Under this convention, the rental SCC is determined by the discount rate applied to the standing stock for one year, not by an assumed short-term SCC growth path.

We use a carbon price of $678.33 per metric ton of carbon (tC). This equals the Rennert et al. (Rennert et al. 2022) mean SCC at a 2% discount rate, converted from CO to carbon units ($185.05/tCO × 3.664 tCO /tC). The resulting annual rental rate is $13.43 per tC in 2020 US dollars. We convert this to 2019 dollars using the World Bank GDP deflator for 2019 (1.85%), giving $13.43 / 1.0185 $13.19 per tC. The same rental price is applied to mangrove, salt marsh and seagrass stocks.

Across the four Ramsey discount rates considered here (1.5%, 2.0%, 2.5%, 3.0%), the inflation-adjusted rental SCC ranges from $8.51 per tC at a 3.0% rate to $16.50 per tC at a 1.5% rate. The 2.0% baseline is $13.19 per tC. Table 1 reports these rental prices.

Table 1: Rental social cost of carbon at alternative Ramsey discount rates.

Discount rate	Growth rate (carbon price)	Net discount rate (discount rate − growth rate)	SC-CO (2020$)	SCC (2020$)	Rental SCC (2020$)	Rental SCC (2019$)
0.015	0	0.015	308	1,129.33	16.81	16.50
0.020	**0**	**0.020**	**185**	**678.33**	**13.43**	**13.19**
0.025	0	0.025	118	432.67	10.68	10.49
0.030	0	0.030	80	293.33	8.67	8.51

3 Results

Coastal blue carbon storage generates approximately $26.65 billion in one-year GEP value in 2019 dollars. The account covers 63.4 million hectares of mapped mangrove, salt marsh and seagrass habitat inside national EEZs. These ecosystems store 2.02 Pg C, valued at the baseline rental SCC of $13.19 per tC.

Mangroves dominate the global value because of their high per-hectare carbon density. Seagrass contributes substantially because of its large mapped extent, despite lower per-hectare stocks. Salt marshes contribute a mid-sized share, mainly from temperate and high-latitude coastlines.

Table 2 shows the details. Mangroves have the highest carbon density and value density. Seagrass contributes strongly to the global total because its mapped area is large, despite lower per-hectare stocks.

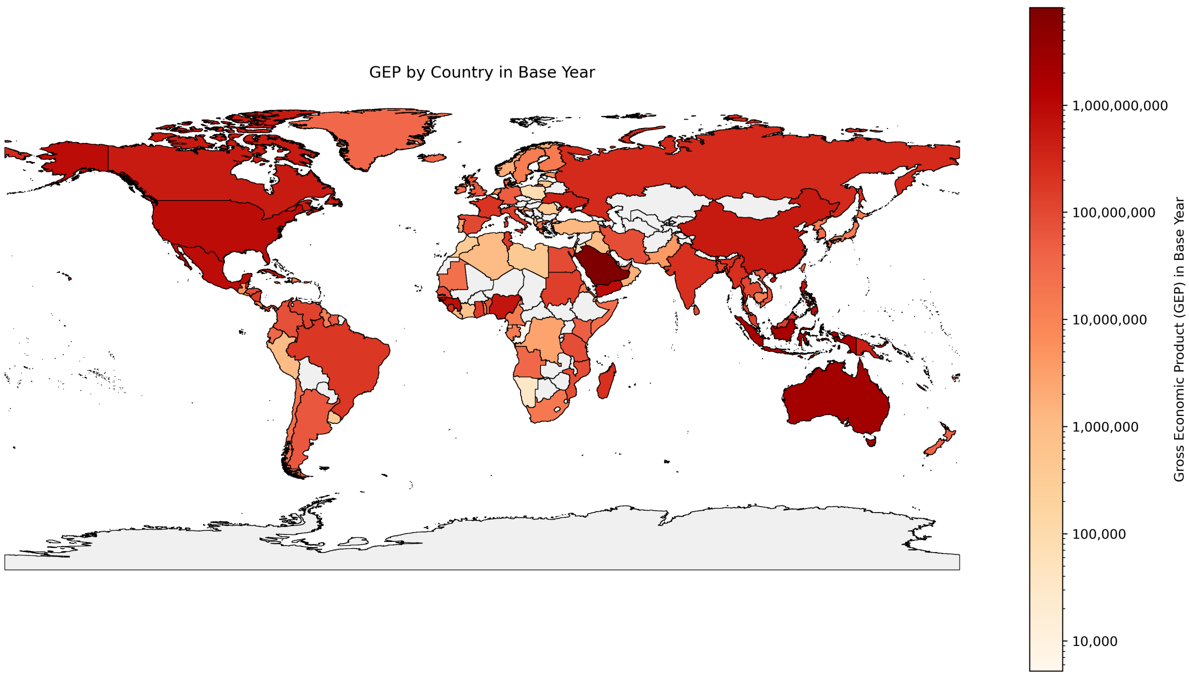
Table 2: Per-hectare coastal carbon stock and annual rental value by ecosystem.

	Ecosystem	Area Mha	Stock Pg C	Value bn 2019$	Stock density ton C/ha	Value density $/ha/yr	Value share %
0	Mangrove	0.37	0.31	4.03	835.80	11,019.44	15.10
1	Salt marsh	0.86	0.20	2.68	237.70	3,134.42	10.10
2	Seagrass	62.21	1.51	19.93	24.30	320.37	74.80

Because the one-year 2019 storage value scales linearly with the rental price, the global value is approximately $17.19 billion at a 3.0% discount rate and $33.33 billion at a 1.5% rate.

The map below shows country-level storage value on a logarithmic colour scale. Countries with large mangrove forests and extensive seagrass beds rank highest. Full country rankings are in the Appendix.

Figure 2: Coastal carbon storage value by country in the base year.



Notes: Country values are shown on a logarithmic colour scale. Countries with no positive mapped coastal carbon storage value are shown in grey.

4 Validation

We validate the coastal-carbon results in two steps. First, we compare the monetary value of the global account with Bertram et al. (Bertram et al. 2021), the closest published global country-level economic assessment of blue carbon wealth. Second, we compare the physical carbon-stock totals with published ecosystem-specific estimates for mangroves, salt marshes and seagrasses.

Bertram et al. (Bertram et al. 2021) estimate global blue carbon wealth from annual carbon sequestration at US\$190.7 $\pm$ 29.5 billion yr^{-1}, using a global mean SCC of US\$640.3 $\pm$ 4.93 tCO_2^{-1}. Our baseline estimate is US\$26.65 billion yr^{-1} in 2019 dollars, using an SC-CO_2 of US\$185.05 tCO_2^{-1} and an annual rental SCC of US\$13.19 tC^{-1}. Our value is lower for two reasons. Bertram et al. apply a higher full SCC to annual sequestration, while this study applies a rental SCC to standing carbon stock. The physical quantity also differs: annual sequestration in Bertram et al., versus stored carbon stock here.

We then compare physical stock estimates by ecosystem. Differences mainly reflect three choices: the extent layer, the 20 cm soil-depth convention, and the carbon-density product. The key density inputs are the Sanderman 2018 raster for mangrove SOC, latitude step functions for salt marsh SOC, and Fourqurean global mean for seagrass SOC.

4.1 Mangrove

The pipeline produces a global mangrove total of 5.52 Pg C over the GMW v3 2019 extent (~14.7 Mha), decomposed in Table 3.

Table 3: Global mangrove carbon stock, this study vs published estimates.

Component	This study (Pg C)	Published (Pg C)	Source
Above-ground biomass C	2.81	1.5 – 3.0	Hamilton & Friess (Hamilton and Friess 2018); Hutchison (Hutchison et al. 2014); Simard (Simard et al. 2019)
Below-ground biomass C	1.15	0.7 – 1.5	IPCC Wetlands Supplement ratios (IPCC 2014)
Soil organic C, top 20 cm	1.56	1.28	Sanderman (Sanderman et al. 2018), scaled to 20 cm
Total ecosystem C	5.52	5.72 ± 0.10 (2020, 13.6 Mha, full depth)	Ju (Ju et al. 2025)

Our mangrove SOC total (1.56 Pg C) is about 22 percent above the Sanderman et al. 20 cm figure of 1.28 Pg C (6.4 Pg C at 1 m scaled by 0.2) (Sanderman et al. 2018). The main reason is extent: all-touched rasterisation of GMW v3 onto the 10 arc-second grid retains thin coastal fringes that are not fully captured in the original 30 m Sanderman product. Our above-ground biomass total (2.81 Pg C) sits at the upper end of the published range and is consistent with applying the Hamilton and Friess latitude regression (Hamilton and Friess 2018) over the broader GMW v3 extent. Our total (5.52 Pg C) is close to the Ju et al. (Ju et al. 2025) standing-stock estimate of 5.72 Pg C for 2020, reflecting the larger GMW v3 extent partially offset by our 20 cm soil depth convention.

4.2 Salt marsh

The pipeline produces a global salt marsh total of 0.90 Pg C over the GLC_FCS30D 2019 extent, decomposed in Table 4.

Table 4: Global salt marsh carbon stock, this study vs published estimates.

Component	This study (Pg C)	Published (Pg C)	Source
Above-ground biomass C	0.038	0.03 – 0.05	Chmura (Chmura et al. 2003)
Below-ground biomass C	0.086	0.07 – 0.15	Howard manual ratios (Howard et al. 2014)

Component	This study (Pg C)	Published (Pg C)	Source
Soil organic C, top 20 cm	0.78	0.28–0.42; 0.08 (McLeod 2011)	Maxwell (Maxwell et al. 2024); McLeod (McLeod et al. 2011)
Total ecosystem C	0.90	1.5 – 2.1 (full depth)	Maxwell (Maxwell et al. 2024)

Our salt marsh SOC total (0.78 Pg C) is about twice the Maxwell et al. (Maxwell et al. 2024) global estimate of 0.28 to 0.42 Pg C for top 20 cm. The main driver is the latitude step function, which assigns 36 to 70 ton C/ha (20 cm) across latitude bands. Maxwell et al. report a global area-weighted mean of ~225 ton C/ha (1 m equivalent), which scales to ~45 ton C/ha at 20 cm depth. Replacing the step function with the native raster from that source is the most important remaining improvement for salt marshes.

4.3 Seagrass

The pipeline produces a global seagrass total of 1.08 Pg C over the WCMC v7.1 extent (~16 – 27 Mha, McKenzie 2020 confidence range), decomposed in Table 5.

Table 5: Global seagrass carbon stock, this study vs published estimates.

Component	This study (Pg C)	Published (Pg C)	Source
Above-ground biomass C	0.046	0.007 – 0.012	Gomis (Gomis et al. 2025)
Below-ground biomass C	0.107	0.017 – 0.028	Gomis (Gomis et al. 2025)
Total biomass C	0.153	0.024 – 0.040	Gomis (Gomis et al. 2025)
Soil organic C, top 20 cm	0.93	0.84–1.68	Fourqurean (Fourqurean et al. 2012); Macreadie (Macreadie et al. 2021)
Total ecosystem C	1.08	4.3 – 8.4 (full depth)	Fourqurean (Fourqurean et al. 2012)

Our seagrass biomass-only total is about four to six times the Gomis et al. 2025 range of 0.024 to 0.040 Pg C (Gomis et al. 2025). This happens because about 78 percent of WCMC polygons lack genus information and receive the Gomis global mean of 1.55 ton C/ha. That value is conservative for individual stocks but generous when applied across the full WCMC extent rather than the McKenzie high-confidence subset. The WCMC v7.1 extent of ~27 Mha is also at the upper end of the published range. Our soil total (0.93 Pg C) uses the Fourqurean et al. 2012 global mean of 22 ton C/ha for the top 20 cm, which is conservative relative to the Fourqurean full-ecosystem estimate of 4.3 to 8.4 Pg C because we apply only the soil component and use a subset of the global extent.

5 Conclusion

This report estimates the one-year 2019 GEP value of standing coastal blue carbon stocks across national EEZs. It combines recent extent and density products for mangroves, salt marshes and seagrasses with the rental social cost of carbon at a 2 percent Ramsey discount rate. The approach extends EEZ-based country attribution from annual sequestration flows (Bertram et al. 2021) to standing-stock storage.

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7 Appendix

7.1 Top countries

	Country	Storage value (2019$)
191	Saudi Arabia	8,160,731,472.07
14	Australia	2,379,048,526.96
102	Indonesia	1,876,579,303.43
87	Guinea-Bissau	947,995,991.65
234	United States	874,160,424.79
246	Yemen	856,563,166.74
142	Mexico	852,143,867.63
84	Guinea	809,164,631.92
176	Philippines	715,144,580.02
163	Nigeria	609,396,783.50

7.2 Coastal Carbon GEP by income group

	income_grp	value
0	1. High income: OECD	4,569,203,137.08
1	2. High income: nonOECD	8,729,212,590.92
2	3. Upper middle income	3,631,483,943.98
3	4. Lower middle income	6,620,381,774.41
4	5. Low income	3,097,362,585.47

7.3 Coastal Carbon GEP by UN region

	region_un	value
0	Africa	4,034,899,762.93
1	Americas	4,248,106,750.02
2	Antarctica	0.00
3	Asia	13,587,929,715.17
4	Europe	1,355,097,834.88
5	Oceania	3,408,244,816.79
6	Seven seas (open ocean)	13,365,152.07

7.4 Coastal Carbon GEP by continent

	continent	value
0	Africa	4,030,208,283.97
1	Antarctica	0.00
2	Asia	13,514,181,368.57
3	Europe	1,355,097,834.88
4	North America	3,677,751,695.44
5	Oceania	3,408,244,816.79

	continent	value
6	Seven seas (open ocean)	78,613,624.43
7	South America	583,546,407.78

7.5 Coastal Carbon GEP by subregion

	subregion	value
0	Antarctica	0.00
1	Australia and New Zealand	2,421,340,149.32
2	Caribbean	851,922,639.52
3	Central America	1,404,089,432.70
4	Central Asia	0.00
5	Eastern Africa	521,849,786.98
6	Eastern Asia	622,334,709.47
7	Eastern Europe	626,158,155.24
8	Melanesia	704,026,517.63
9	Micronesia	108,484,200.51
10	Middle Africa	65,744,679.00
11	Northern Africa	395,415,437.49
12	Northern America	1,408,548,270.02
13	Northern Europe	195,269,905.24
14	Polynesia	174,393,949.32
15	Seven seas (open ocean)	13,365,152.07
16	South America	583,546,407.78
17	South-Eastern Asia	3,179,443,516.17
18	Southern Africa	16,484,890.27
19	Southern Asia	612,445,324.84
20	Southern Europe	274,881,849.16
21	Western Africa	3,035,404,969.20
22	Western Asia	9,173,706,164.70
23	Western Europe	258,787,925.24

7.6 Coastal Carbon GEP by Country in Base Year

	iso3_r250_name	value
0	Aruba	1,332,397.36
1	Afghanistan	0.00
2	Angola	33,104,135.45
3	Anguilla	280,252.81
4	Åland	0.00
5	Albania	1,450,541.08
6	Andorra	0.00
7	United Arab Emirates	59,911,524.06

	iso3_r250_name	value
8	Argentina	59,332,327.64
9	Armenia	0.00
10	American Samoa	13,515,592.75
11	Antarctica	0.00
12	French Southern Territories	173,798.86
13	Antigua and Barbuda	7,985,745.10
14	Australia	2,379,048,526.96
15	Austria	0.00
16	Azerbaijan	0.00
17	Burundi	0.00
18	Belgium	6,101.26
19	Benin	78,988,120.62
20	Bonaire, Sint Eustatius and Saba	0.00
21	Burkina Faso	0.00
22	Bangladesh	122,710,487.66
23	Bulgaria	0.00
24	Bahrain	13,299,381.84
25	Bahamas	135,340,922.79
26	Bosnia and Herzegovina	0.00
27	Saint Barthélemy	0.00
28	Belarus	0.00
29	Belize	177,749,358.74
30	Bermuda	16,041.71
31	Bolivia	0.00
32	Brazil	175,797,975.66
33	Barbados	2,227,708.12
34	Brunei	3,697,900.57
35	Bhutan	0.00
36	Bouvet Island	0.00
37	Botswana	0.00
38	Central African Republic	0.00
39	Canada	500,859,212.76
40	Cocos Islands	0.00
41	Switzerland	0.00
42	Chile	23,818,828.99
43	China	507,495,579.14
44	Côte d'Ivoire	564,490.30
45	Cameroon	14,814,902.49
46	Democratic Republic of the Congo	2,681,708.47
47	Republic of the Congo	5,597.45
48	Cook Islands	62,460.17
49	Colombia	76,541,264.05
50	Comoros	38,002,961.62
51	Cabo Verde	0.00
52	Costa Rica	9,615,742.71
53	Cuba	581,958,640.64

	iso3_r250_name	value
54	Curaçao	0.00
55	Christmas Island	0.00
56	Cayman Islands	1,437,311.46
57	Cyprus	2,127,062.79
58	Czechia	0.00
59	Germany	49,787,941.69
60	Djibouti	660,832.66
61	Dominica	14,517,521.62
62	Denmark	37,128,635.27
63	Dominican Republic	13,093,893.93
64	Algeria	1,054,602.03
65	Ecuador	35,179,745.13
66	Egypt	91,033,527.05
67	Eritrea	12,485,744.63
68	Western Sahara	0.00
69	Spain	98,467,485.81
70	Estonia	5,197,382.81
71	Ethiopia	0.00
72	Finland	14,647,646.57
73	Fiji	63,161,219.20
74	Falkland Islands	16,439,651.24
75	France	202,633,660.65
76	Faroe Islands	391,185.07
77	Micronesia	54,487,558.33
78	Gabon	14,806,476.20
79	United Kingdom	63,605,533.77
80	Georgia	0.00
81	Guernsey	469,366.47
82	Ghana	158,271,679.90
83	Gibraltar	0.00
84	Guinea	809,164,631.92
85	Guadeloupe	0.00
86	Gambia	14,429,398.17
87	Guinea-Bissau	947,995,991.65
88	Equatorial Guinea	331,858.94
89	Greece	20,190,388.80
90	Grenada	568,377.87
91	Greenland	33,386,097.79
92	Guatemala	4,841,263.81
93	French Guiana	0.00
94	Guam	13,193,948.58
95	Guyana	13,554,180.85
96	China, Hong Kong Special Administrative Region	21,713,394.57
97	Heard Island and McDonald Island	0.00
98	Honduras	87,229,469.75
99	Croatia	7,687,234.07

	iso3_r250_name	value
100	Haiti	0.00
101	Hungary	0.00
102	Indonesia	1,876,579,303.43
103	Isle of Man	0.00
104	India	229,493,221.06
105	British Indian Ocean Territory	0.00
106	Ireland	22,245,559.44
107	Iran	82,512,780.99
108	Iraq	944,876.26
109	Iceland	33,831,720.56
110	Israel	206,850.32
111	Italy	140,752,220.99
112	Jamaica	26,210,767.64
113	Jersey	2,048,350.17
114	Jordan	47,187.96
115	Japan	15,212,571.76
116	Kazakhstan	0.00
117	Kenya	48,041,427.92
118	Kyrgyzstan	0.00
119	Cambodia	12,582,793.57
120	Kiribati	0.00
121	Saint Kitts and Nevis	1,674,507.26
122	South Korea	26,831,458.83
123	Kuwait	45,178,395.98
124	Laos	0.00
125	Lebanon	0.00
126	Liberia	1,136,243.34
127	Libya	414,227.84
128	Saint Lucia	312,461.64
129	Liechtenstein	0.00
130	Sri Lanka	99,878,449.13
131	Lesotho	0.00
132	Lithuania	411,461.50
133	Luxembourg	0.00
134	Latvia	5,227.72
135	China, Macao Special Administrative Region	0.00
136	Saint-Martin	1,917,775.70
137	Morocco	335,760.45
138	Monaco	16,327.88
139	Moldova	0.00
140	Madagascar	221,514,960.00
141	Maldives	73,748,346.61
142	Mexico	852,143,867.63
143	Marshall Islands	14,715,408.06
144	North Macedonia	0.00
145	Mali	0.00

	iso3_r250_name	value
146	Malta	3,697,093.69
147	Myanmar	240,213,637.68
148	Montenegro	0.00
149	Mongolia	0.00
150	Northern Mariana Islands	0.00
151	Mozambique	85,214,127.71
152	Mauritania	23,636,476.69
153	Montserrat	390,226.77
154	Martinique	0.00
155	Mauritius	3,960,255.36
156	Malawi	0.00
157	Malaysia	136,939,836.93
158	Mayotte	0.00
159	Namibia	34,563.92
160	New Caledonia	56,410,104.98
161	Niger	0.00
162	Norfolk Island	0.00
163	Nigeria	609,396,783.50
164	Nicaragua	171,398,391.69
165	Niue	0.00
166	Netherlands	6,343,893.75
167	Norway	1,826,764.36
168	Nepal	0.00
169	Nauru	0.00
170	New Zealand	42,291,622.36
171	Oman	1,642,628.26
172	Pakistan	4,102,039.38
173	Panama	96,997,622.21
174	Pitcairn Islands	0.00
175	Peru	1,070,453.84
176	Philippines	715,144,580.02
177	Palau	26,087,285.54
178	Papua New Guinea	424,379,001.93
179	Poland	80,440.26
180	Puerto Rico	23,594,760.10
181	North Korea	13,214,901.24
182	Portugal	2,630,046.57
183	Paraguay	0.00
184	Palestine	0.00
185	French Polynesia	20,279,865.14
186	Qatar	31,879,837.87
187	Réunion	0.00
188	Romania	426,106.01
189	Russia	280,520,656.76
190	Rwanda	0.00
191	Saudi Arabia	8,160,731,472.07

	iso3_r250_name	value
192	Sudan	133,939,328.27
193	Senegal	90,901,787.20
194	Singapore	7,888,723.59
195	South Georgia and the South Sand	0.00
196	Saint Helena, Ascension and Tris	0.00
197	Svalbard and Jan Mayen	0.00
198	Solomon Islands	125,002,699.97
199	Sierra Leone	289,538,250.81
200	El Salvador	4,113,716.16
201	San Marino	0.00
202	Somalia	27,426,452.82
203	Saint Pierre and Miquelon	126,492.97
204	Serbia	0.00
205	South Sudan	0.00
206	São Tomé and Príncipe	0.00
207	Suriname	59,676,490.72
208	Slovakia	0.00
209	Slovenia	6,838.16
210	Sweden	13,461,071.52
211	Swaziland	0.00
212	Sint Maarten	1,408,503.74
213	Seychelles	731,223.60
214	Syria	0.00
215	Turks and Caicos Islands	33,464,280.91
216	Chad	0.00
217	Togo	11,381,115.10
218	Thailand	111,831,126.69
219	Tajikistan	0.00
220	Tokelau	0.00
221	Turkmenistan	0.00
222	Timor-Leste	11,824,713.86
223	Tonga	111,352,036.46
224	Trinidad and Tobago	2,776,024.59
225	Tunisia	168,637,991.85
226	Turkey	1,173,780.56
227	Tuvalu	27,675.95
228	Taiwan	37,866,803.93
229	Tanzania	83,811,800.65
230	Uganda	0.00
231	Ukraine	345,130,952.22
232	United States Minor Outlying Isl	13,191,353.21
233	Uruguay	395,476.34
234	United States	874,160,424.79
235	Uzbekistan	0.00
236	Vatican City	0.00
237	Saint Vincent and the Grenadines	121,073.64

	iso3_r250_name	value
238	Venezuela	121,740,013.31
239	British Virgin Islands	1,309,485.80
240	Virgin Islands, U.S.	0.00
241	Vietnam	62,740,899.83
242	Vanuatu	35,073,491.55
243	Wallis and Futuna	225,739.60
244	Samoa	28,930,579.25
245	Serbia	0.00
246	Yemen	856,563,166.74
247	South Africa	16,450,326.35
248	Zambia	0.00
249	Zimbabwe	0.00